



Smart Visual Crop Advisory & Disease Severity Estimation System

Multi-Modal Deep Learning, Explainable AI (Grad-CAM), Microclimate Telemetry & Vernacular Voice Prescriptions

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I. Problem Statement

 **1. Severe Yield Loss & Delayed Diagnosis:** Crop diseases destroy 20–40% of yields annually. Farmers lack on-field access to agricultural extension scientists during critical infection windows.

 **2. Black-Box AI Trust Deficit:** Existing deep learning models output opaque labels without visual localization, causing farmer skepticism and misdiagnosis.

 **3. Meteorological Disconnection:** Models diagnose leaves in isolation without factoring in ambient humidity ($\geq 75\%$) and rainfall wetness that drive fungal multiplication.

 **4. Language & Practicality Barrier:** Text-heavy English outputs fail illiterate farmers and omit practical 15-liter backpack spray pump dosages.

II. Objectives

 **1. High-Accuracy Edge Deep Learning:** Deploy MobileNetV2 classifying 39 crop taxa at 98.43% accuracy with <30 ms latency.

 **2. Visual Explainability (Grad-CAM):** Generate gradient heatmaps visually highlighting active fungal necrotic lesions.

 **3. Real-Time Microclimate Risk Engine:** Ingest Open-Meteo GPS weather to compute Disease Triangle risk score (0–100).

 **4. Multilingual Zero-Literacy Voice Assistant:** Build interactive Web Speech voice assistant in Kannada, Hindi, and English.

 **5. Instant Rural Deliverables:** 1-Click printable PDF health prescription card and 1-tap WhatsApp sharing for local dealers.

III. Literature Survey

Journal / Year / Authors	Methodology	Key Findings	Research Gap & Our Solution
Scientific Reports (Nature), 2024 M. J. Karim, M. O. F. Goni, Md. Nahiduzzaman, M. Ahsan, J. Haider, M. Kowalski	• Custom MobileNetV3Large with dense dropout layers. • Grad-CAM visual heatmaps for lesion grounding. • Deployed on Nvidia Jetson Nano edge device. • Dataset: 27,122 grape leaf images (4 classes).	• Achieved 99.42% accuracy, precision, and recall. • Proved feasibility of real-time edge XAI inference.	• Gap: Tested only on single crop (grape) with no weather telemetry, multilingual voice assistance, or field spray pump dosages. • Our Solution: Multi-crop (39 classes) + Open-Meteo weather risk + Kannada/Hindi voice.
IEEE Access, 2023 Md. S. H. Shovon, S. K. Mozumder, O. Saha, M. F. Hossain, M. Z. Islam	• PlantDet: Multi-model ensemble architecture (dResNet-18, InceptionV3, DenseNet). • Trained on 90,000 multi-crop images (19 classes).	• Achieved 99.10% overall classification accuracy across diverse plant species.	• Gap: High computational footprint of multi-model ensembles hinders mobile edge deployment; lacks XAI heatmaps. • Our Solution: Lightweight MobileNetV2 (2.2M params, 22ms) with instant Grad-CAM visual proof.
Artificial Intelligence in Agriculture (Elsevier), 2022 A. S. Paymode and V. B. Malode	• VGG-16 deep CNN transfer learning architecture. • Evaluated on 62,286 multi-crop leaf images (14 classes: Grapes & Tomatoes).	• Achieved 98.40% classification accuracy on dual-crop disease diagnosis.	• Gap: 138M parameter model causes high latency; omitted precision/recall/F1 and provided no actionable farm dosages. • Our Solution: Calibrated 30g/15L pump spray dosages + organic remedies with top-3 confidence.
Frontiers in Plant Science, 2021 P. Wang, E. Niu, Y. Zheng, Y. Chen, F. Yuan	• ECA-sNET (Efficient Channel Attention + lightweight CNN). • Evaluated on 6,867 grape leaf images across 6 classes.	• Achieved 99.66% accuracy, 99.66% precision, and 99.60% F1-score on fine-grained lesion features.	• Gap: Evaluated solely on isolated dataset without microclimatic context or vernacular accessibility for farmers. • Our Solution: Real-time weather triangulation + hands-free Kannada, Hindi & English oral interaction.
Frontiers in Plant Science, 2016 S. P. Mohanty, D. P. Hughes, and M. Salathé	• Landmark benchmark study using AlexNet & GoogLeNet CNNs. • PlantVillage dataset of 54,306 images across 38 crop disease classes.	• Achieved 99.35% accuracy on studio dataset, establishing the standard plant disease benchmark.	• Gap: Severe accuracy drop on real-world farm conditions (domain shift); black-box with no visual grounding or localized advice. • Our Solution: Out-of-Distribution thresholding (<60%) + Grad-CAM grounding + WhatsApp share.

IV. Analysis: Existing System vs. Proposed System

EXISTING SYSTEM

- **Manual Field Scouting:** Slow, labor-intensive, error-prone, and dependent on scarce agricultural scientists.
- **Black-Box AI Models:** Output opaque prediction labels with zero visual confirmation of which leaf lesions triggered the diagnosis.
- **Isolated Vision (No Weather):** Diagnoses image without checking today's humidity (82%) or rain, failing to forecast disease multiplication.
- **Text-Heavy English Jargon:** Inaccessible to illiterate or vernacular-speaking rural farmers.
- **Vague Chemical Advice:** Mentions chemical names without practical 15-liter backpack pump spray dosages.

PROPOSED: AGROVOICE-XAI

- **Lightweight Edge MobileNetV2:** 98.43% validation accuracy, 2.2M parameters, and 22 ms inference latency across 39 crop classes.
- **Active Grad-CAM Saliency (XAI):** Generates visual glowing red/yellow heatmaps over active fungal necrotic lesions for 100% farmer trust.
- **Live Weather Triangulation:** Open-Meteo GPS weather telemetry computes live Disease Triangle spread risk score (0–100).
- **Zero-Literacy Vernacular Voice:** Interactive hands-free speech recognition (STT) and voice synthesis (TTS) in Kannada, Hindi, and English.
- **4-Pillar Calibrated Prescriptions:** Organic remedies + exact 30g/15L pump chemical spray dosages + 1-click WhatsApp and PDF share.

V. Requirements Specification (Functional & Non-Functional)

Functional Requirements (FR)

- **FR-1:** Image acquisition & camera framing guide.
- **FR-2:** MobileNetV2 39-class disease classification.
- **FR-3:** Grad-CAM visual heatmap synthesis.
- **FR-4:** Live Open-Meteo GPS weather ingestion.
- **FR-5:** Web Speech voice assistant in KN/HI/EN.
- **FR-6:** Printable PDF card & WhatsApp share.

Non-Functional Requirements (NFR)

- **NFR-1 (Latency):** Model inference < 30 ms; pipeline < 350 ms.
- **NFR-2 (Accuracy):** 98.43% accuracy; 0.982 macro F1-score.
- **NFR-3 (Safety):** Out-of-distribution (<60%) safety threshold.
- **NFR-4 (Usability):** Zero-literacy voice & high contrast UI.
- **NFR-5 (Modularity):** Asynchronous FastAPI architecture.

VI. System Architecture & Data Flow Design

1. Client Edge Layer (React 18 + Vite): Camera viewfinder, aspect ratio alignment, Web Speech API (STT/TTS).

2. FastAPI Gateway: Asynchronous payload decoding, tensor normalization (224x224x3 ImageNet).

3. Parallel Deep Learning & Weather: MobileNetV2 (98.43%) + Grad-CAM hook at features[-1] + Open-Meteo Disease Triangle risk engine.

4. 4-Pillar Agronomic Knowledge Engine: Etiology, Organic remedies, 30g/15L pump chemical dosage, Cultural prevention.

5. Deliverables: Side-by-side XAI view, Kannada/Hindi audio playback, 1-click PDF card & WhatsApp share.

VII. Summary & Experimental Results

 **98.43% Validation Accuracy:** Validated across 54,305 curated leaf images covering 39 distinct crop taxa.

 **22 ms Edge Latency:** Lightweight inverted residual MobileNetV2 runs in real-time on standard CPUs/smartphones.

 **100% Explainable AI Localization:** Grad-CAM successfully pinpoints necrotic lesion centers on diseased leaves.

 **Microclimatic Risk Triangulation:** Accurately predicts high fungal multiplication risk during high humidity (>80% RH).

 **Zero-Literacy Rural Impact:** Spoken advice in Kannada, Hindi, and English with standardized 15L pump spray dosages.

VIII. Publication & Conference Details

 Paper Title:

AgroVoice-XAI: An Explainable Multimodal Deep Learning Framework for Crop Disease Diagnosis Augmented with Microclimatic Risk Telemetry and Vernacular Voice Assistance

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 **Target Conference:** IEEE International Conference on Computing, Communication and Automation (ICCCA) / Springer Smart Agricultural Technology & Computing Series

IX. References

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